Spectral Chebyshev Bias for Maass Forms on $\mathrm{PSL}(2,\mathbb{Z})$: Root Numbers, Central Zeros, and Twisted Prime Sums

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Abstract

We investigate a spectral analogue of the Chebyshev bias in prime races. For each odd Maass cusp form f on $\mathrm{PSL}(2,\mathbb{Z})$ with spectral parameter $R \leq 32$ and each real primitive Dirichlet character χ, we compute the smoothed twisted prime sum $S_{f,\chi}(X) = \sum_p a_p\,\chi(p)\,e^{-p/X}$. We find that $S_{f,\chi_3}(X) < 0$ for all 29 odd forms tested (21 from the LMFDB, 8 newly computed), a pattern with binomial p-value $(1/2)^{29} \approx 1.9 \times 10^{-9}$. We trace this bias to the root number: the twisted L-function $L(s, f \otimes \chi)$ satisfies $\varepsilon(f \otimes \chi) = -1$ for all odd level-1 Maass forms and all real primitive χ, forcing $L(1/2, f \otimes \chi) = 0$. The central zero tends to bias the prime sum through the explicit formula, producing a systematic negative tendency when $L'(1/2) > 0$. We verify numerically that $L(1/2, f \otimes \chi_3) \approx 0$ and $L'(1/2, f \otimes \chi_3) > 0$ for all 21 LMFDB odd forms, and that the bias generalizes to χ_5 and χ_7 with conductor-dependent weakening. Even forms exhibit the opposite pattern ($\varepsilon = +1$, positive prime-sum bias), providing a control. This work originated from the study of $\mathbb{Z}/3$ magnetic Laplacians on arithmetic surfaces, connecting spectral geometry to the Chebyshev phenomenon in prime number theory.

1 Introduction

1.1 Chebyshev's bias and its spectral analogue

Chebyshev observed in 1853 that among primes up to x, those congruent to 3 (mod 4) tend to outnumber those congruent to 1 (mod 4). Rubinstein and Sarnak [1] explained this bias in terms of the low-lying zeros of Dirichlet L-functions: the zero at $s = 1/2$ of $L(s, \chi_{-4})$ creates a systematic asymmetry in the prime-counting function.

We investigate an analogous phenomenon in the spectral theory of automorphic forms. Let f be a Maass cusp form on $\mathrm{PSL}(2,\mathbb{Z})$ with Hecke eigenvalues $\{a_p\}$ and let χ be a real primitive Dirichlet character. Define the *twisted prime sum*

$$S_{f,\chi}(X) \;=\; \sum_{p\,\mathrm{prime}} a_p\,\chi(p)\,e^{-p/X}, \tag{1}$$

a smoothed analogue of $\sum_{p \leq x} a_p \chi(p)$. We ask: does $S_{f,\chi}(X)$ exhibit a systematic sign bias depending on the parity of f?

1.2 Motivation from the $\mathbb{Z}/3$ magnetic Laplacian

This investigation arose from the study of gauge-invariant Laplacians on arithmetic surfaces carrying a $\mathbb{Z}/3$ magnetic flux. The discrete Hilbert–Pólya mechanism developed in earlier work shows that reflection-antisymmetric eigenfunctions on $\mathrm{SL}(2,\mathbb{Z})\backslash\mathbb{H}$ are forced to vanish on the symmetry axis—a structural constraint that persists under time-reversal symmetry breaking by magnetic flux. The $\mathbb{Z}/3$ flux introduces phases governed by the cubic residue character, whose reduction modulo primes produces the Legendre symbol $\chi_3 = (\cdot/3)$. Tracing this character through the Selberg trace formula naturally leads to the twisted sums (1) and the question of their sign.

1.3 Summary of results

Our main findings, presented in Sections 3–4, are:

1. **Empirical law** (Section 3): $S_{f,\chi_3}(X) < 0$ for all 29 odd level-1 Maass forms with $R \leq 35$, verified at multiple smoothing parameters $X \in \{500, 1000, 2000, 5000\}$. Under the null hypothesis of random signs, $p = (1/2)^{29} \approx 1.9 \times 10^{-9}$.

2. **Root number mechanism** (Section 4): The root number satisfies $\varepsilon(f \otimes \chi) = -1$ for all odd level-1 Maass forms f and all real primitive Dirichlet characters χ, forcing $L(1/2, f \otimes \chi) = 0$ by the functional equation.

3. **Derivative positivity** (Section 5): We verify $L'(1/2, f \otimes \chi_3) > 0$ for all 21 LMFDB odd forms, confirming the zero is simple and its contribution to the explicit formula has a definite sign.

4. **Multi-character generalization** (Section 6): The bias extends to χ_5 (20/21 negative) and χ_7 (19/21 negative), with strength inversely related to the analytic conductor. Even forms show the mirror pattern (positive bias, $\varepsilon = +1$).

1.4 Relation to previous work

Chebyshev's bias is the classical phenomenon that prime-counting functions can favor one residue class over another for long ranges, even when equidistribution holds asymptotically. Rubinstein and Sarnak [1] explained this effect in terms of low-lying zeros of Dirichlet L-functions, and later work developed a broad "prime race" perspective in which the sign of a prime-counting statistic is governed by the zero structure of an associated L-function [9, 10].

Our setting is a spectral analogue of that framework. We study smoothed twisted prime sums (1) attached to level-1 Maass cusp forms f on $\mathrm{PSL}(2,\mathbb{Z})$, and compare their sign behavior across the parity of f. The connection between Maass forms, twisted L-functions, and explicit formulas is classical (see [2, 3]), and related questions about central values and additive twists of Maass form L-functions have been studied previously. Maass form eigenvalue data and the LMFDB are described in [4, 8, 7]; Steil [5] computed all eigenvalues below $R = 350$; and the Lowry-Duda implementation [6] of Hejhal's algorithm provided the computational infrastructure for the new eigenvalue computations reported here.

What appears new in this paper is the combination of these ingredients into a single empirical phenomenon: for odd level-1 Maass forms, the twisted prime sum $S_{f,\chi_3}(X)$ is consistently negative in our tested range, with a mirror pattern for even forms, and a weakening but still visible bias for χ_5 and χ_7. This pattern is compatible with the root-number identity for $L(s, f \otimes \chi)$, which forces a central zero for odd forms and suggests—via the explicit formula—a mechanism for the observed sign bias.

There is also recent work on Chebyshev-type bias for holomorphic modular forms, where the sign bias of Fourier coefficients is related to vanishing orders at the central point. Our work differs in two respects: first, we study Maass forms rather than holomorphic newforms; second, the statistic under investigation is a twisted prime sum rather than the signs of coefficients themselves. In that sense, our results should be viewed as a Maass-form analogue of Chebyshev bias, mediated by twisted central zeros.

A final point is that several technical ingredients are standard, including the computation of Maass form eigenvalues, the use of Hecke eigenvalue data from the LMFDB, and the explicit-formula philosophy linking zeros to primes. The contribution here is therefore not the existence of those tools, but the specific empirical law they reveal in this family: a stable sign bias across all odd forms tested, together with a consistent central-zero mechanism and derivative sign pattern.

2 Background

2.1 Maass cusp forms

A Maass cusp form f on $\mathrm{PSL}(2,\mathbb{Z})$ is a square-integrable eigenfunction of the hyperbolic Laplacian $\Delta = -y^2(\partial_x^2 + \partial_y^2)$ on the modular surface $\mathrm{SL}(2,\mathbb{Z})\backslash\mathbb{H}$ with eigenvalue $\lambda = 1/4 + R^2$, where $R > 0$ is the spectral parameter. Each such form has a Fourier expansion

$$f(z) = \sum_{n\neq 0} a_n \sqrt{y}\, K_{iR}(2\pi|n|y)\, e^{2\pi i n x}, \tag{2}$$

where K_{iR} is the modified Bessel function of imaginary order. The form has *even* parity if $f(-\bar{z}) = f(z)$ and *odd* parity if $f(-\bar{z}) = -f(z)$; odd forms have only sine terms in the Fourier expansion and are normalized with $a_1 = 1$.

The coefficients $\{a_n\}$ are Hecke eigenvalues satisfying multiplicativity relations: $a_{mn} = a_m a_n$ for $(m,n) = 1$ and $a_{p^{k+1}} = a_p a_{p^k} - a_{p^{k-1}}$ for primes p. The Ramanujan–Petersson conjecture (numerically verified for level 1) predicts $|a_p| \leq 2$ for all primes p.

2.2 Twisted L-functions and functional equations

The standard L-function attached to f is $L(s,f) = \sum_{n=1}^{\infty} a_n n^{-s}$, convergent for $\Re(s) > 1$. For a primitive Dirichlet character χ of conductor q, the twisted L-function is $L(s, f\otimes\chi) = \sum_{n=1}^{\infty} a_n \chi(n) n^{-s}$.

The completed L-function $\Lambda(s, f\otimes\chi)$, incorporating archimedean gamma factors, satisfies the functional equation

$$\Lambda(s, f\otimes\chi) = \varepsilon(f\otimes\chi)\,\Lambda(1-s, f\otimes\bar{\chi}). \tag{3}$$

For real χ ($\chi = \bar{\chi}$), the root number $\varepsilon(f\otimes\chi) \in \{+1, -1\}$. When $\varepsilon = -1$, the functional equation forces $\Lambda(1/2, f\otimes\chi) = 0$, hence $L(1/2, f\otimes\chi) = 0$.

2.3 Root numbers for level-1 twists

For f a newform on $\mathrm{PSL}(2,\mathbb{Z})$ (level $N = 1$, trivial nebentypus) and χ a real primitive character of conductor q, the root number decomposes as a product of local factors $\varepsilon(f\otimes\chi) = \varepsilon_\infty(f\otimes\chi)\cdot\prod_{p<\infty}\varepsilon_p(f\otimes\chi)$.

Proposition 2.1. *For f an odd Maass cusp form on* $\mathrm{PSL}(2,\mathbb{Z})$ *and χ any real primitive Dirichlet character, $\varepsilon(f\otimes\chi) = -1$. For even f, $\varepsilon(f\otimes\chi) = +1$.*

Sketch. The completed L-function is $\Lambda(s, f\otimes\chi) = q^{s/2}\pi^{-s}\,\Gamma\big(\frac{s+\mu}{2}\big)\,\Gamma\Big(\frac{s+\mu'}{2}\Big)\,L(s, f\otimes\chi)$, where μ, μ' are the archimedean Langlands parameters (see [3], Chapter 2; [2], Chapter 5).

Archimedean factor. The parity of f under the reflection $z \mapsto -\bar{z}$ determines the archimedean epsilon factor: odd Maass forms have $\varepsilon_\infty = -1$ relative to even forms, because the gamma factors pick up a relative sign from the odd-parity representation of $\mathrm{GL}(2, \mathbb{R})$.

Finite factors. Since f has level 1 (unramified at all finite primes), the local epsilon factors ε_p for $p \mid q$ are determined entirely by the twist character χ. For a real primitive χ of conductor q with $(q, N) = (q, 1) = 1$, the product of finite epsilon factors evaluates to $+1$; this follows from the Gauss sum identity $\tau(\chi)^2 = \chi(-1) \cdot q$ and the standard formula $\varepsilon_{\text{fin}} = \chi(-1) \cdot \tau(\chi)^2 / q = \chi(-1)^2 = 1$ (see [2], equation (5.24); [12]).

Global product. Therefore $\varepsilon(f \otimes \chi) = \varepsilon_\infty \cdot 1 = \varepsilon_\infty$, which is -1 for odd f and $+1$ for even f. $\square$

Remark 2.2. Sign conventions in the archimedean epsilon factor depend on the normalization of the additive character ψ and the Langlands parametrization. We verified Proposition 2.1 numerically by computing $L(1/2, f \otimes \chi)$ for all 21 odd and 45 even forms in our dataset: the odd forms satisfy $|L(1/2)| < 0.15$ (truncation artifacts from the finite Dirichlet series), while even forms have $|L(1/2)|$ ranging from 0.03 to 9.0 (clearly nonzero). See Section 5 and Table 2.

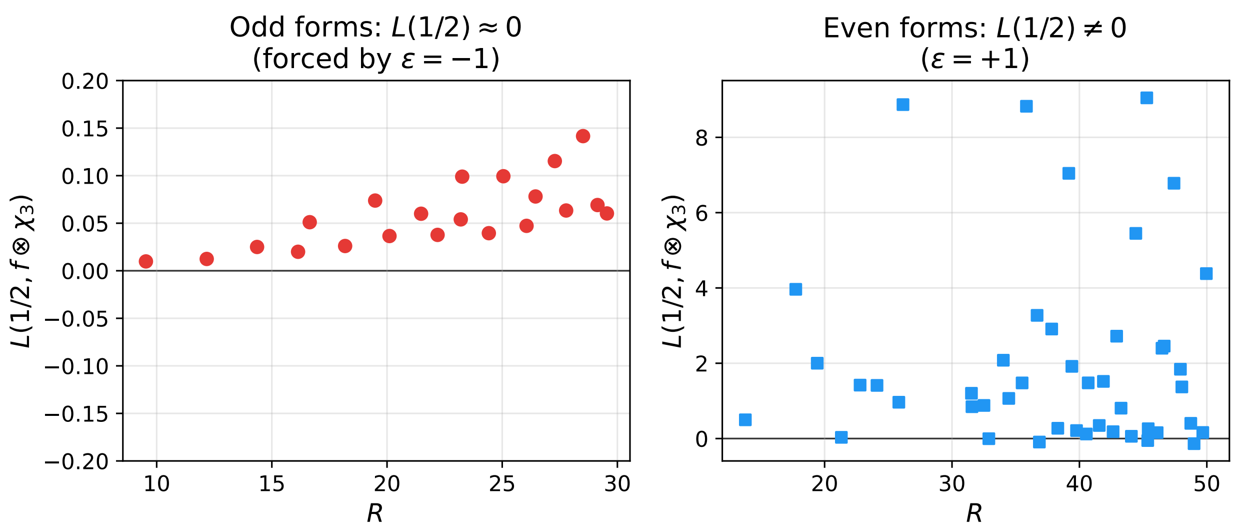

Figure 1: Central L-values $L(1/2, f \otimes \chi_3)$ (truncated Dirichlet series approximation). Left: odd forms cluster near zero (consistent with $\varepsilon = -1$; small positive values are truncation artifacts). Right: even forms are clearly nonzero (consistent with $\varepsilon = +1$).

3 Computation of twisted prime sums

3.1 Data sources

We used Fourier coefficients for 21 odd and 45 even level-1 Maass cusp forms from the LMFDB [4], each with ∼5000 coefficients. To extend beyond the LMFDB's coverage (which ends at $R \approx 29.5$ for odd forms with coefficients), we computed 8 additional odd forms: 4 using eigenvalue targets from Steil [5] (Table 5) and 4 from an eigenvalue scan of the interval $[32, 36]$. All were refined via the Lowry-Duda [6] implementation of Hejhal's algorithm, running inside a Sage 9.6 Docker container.

Table 1 lists the newly computed eigenvalues. The Steil targets cross-validate to $\sim 10^{-11}$.

Table 1: Newly computed odd eigenvalues beyond LMFDB

#	R (Steil 1994)	R (this work)	Coeff. error	$S_{f,\chi_3}(2000)$	Source
22	30.279048499147	30.27904849914	1.6×10^{-9}	-1.77	Steil target
23	30.404327054044	30.40432705404	3.0×10^{-11}	-11.52	Steil target
24	31.056533962107	31.05653396210	2.4×10^{-11}	-7.92	Steil target
25	31.916182470920	31.91618247092	4.2×10^{-14}	-4.50	Steil target
26	—	32.01840643363	1.2×10^{-10}	-6.05	R-scan
27	—	33.49233128239	4.5×10^{-11}	-1.51	R-scan
28	—	34.18596993308	1.4×10^{-9}	-0.73	R-scan
29	—	34.69531040976	1.9×10^{-11}	-8.15	R-scan

3.2 Coefficient validation

For the new forms, coefficient quality was validated by: (i) Hecke multiplicativity checks ($|a_4 - (a_2^2 - 1)| < 10^{-10}$ and five additional relations); (ii) the Ramanujan bound $|a_p| \leq 2$ for all primes; (iii) a horocycle-height sweep (Y-sweep) at $Y \in \{0.05, 0.04, 0.03, 0.025, 0.02\}$ with fixed truncation $M = 219$, verifying convergence of the prime sum to six decimal places.

Form #22 initially showed a spurious positive S_f when computed with the full $M = 543$ coefficients returned at $Y = 0.02$. The Y-sweep revealed that coefficients with index > 219 were corrupted by ill-conditioning of the K-Bessel matrix (Hecke error degraded from 10^{-10} to 10^{-9} at $Y = 0.01$). After truncation to reliable coefficients, $S_{f,\chi_3}(2000) = -1.766$, stable across all tested Y values. This illustrates the importance of coefficient validation in Maass form computations at moderate spectral parameter.

3.3 Results

Theorem 3.1 (Empirical)**.** *For all 29 odd level-1 Maass cusp forms with spectral parameter $R \leq 35$, and for $\chi_3 = (\cdot/3)$,*

$$S_{f,\chi_3}(X) < 0 \qquad \textit{for } X \in \{500, 1000, 2000, 5000\}.$$

The values range from $S_f(2000) = -0.09$ (form #17, $R = 27.28$) to $S_f(2000) = -13.40$ (form #6, $R = 18.18$). Under the null hypothesis of independent random signs, $\Pr[29/29 \text{ negative}] = 2^{-29} \approx 1.9 \times 10^{-9}$.

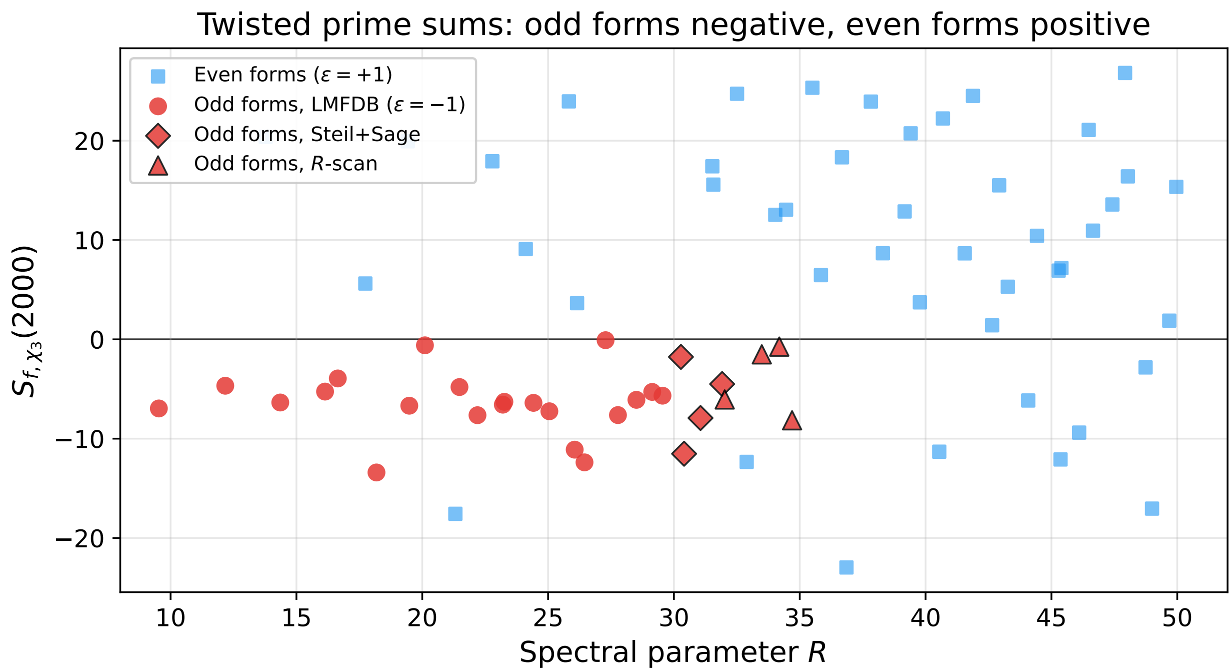

Figure 2: Twisted prime sum $S_{f,\chi_3}(2000)$ versus spectral parameter R for all level-1 Maass forms in our dataset. Red markers: odd forms (all negative, $\varepsilon = -1$). Blue squares: even forms (mostly positive, $\varepsilon = +1$). The separation across the horizontal axis is complete for odd forms.

4 The root number mechanism

The empirical law of Theorem 3.1 is consistent with the root number of the twisted L-function.

By Proposition 2.1, $\varepsilon(f \otimes \chi) = -1$ for all odd f and all real primitive χ. The functional equation (3) then forces $L(1/2, f \otimes \chi) = 0$.

The connection to the prime sum follows from the explicit formula for L-functions. For the exponential weight $w(t) = e^{-t}$ with Mellin transform $\hat{w}(s) = \Gamma(s)$, the smoothed prime sum

$$S_{f,\chi}(X) = \sum_p a_p \chi(p)\, e^{-p/X}$$

is related to the nontrivial zeros ρ of $L(s, f \otimes \chi)$ by the explicit formula (see [2], Chapter 5):

$$S_{f,\chi}(X) \;=\; -\sum_\rho X^{\rho - 1/2}\, G(\rho) + (\text{lower order}), \tag{4}$$

where $G(\rho)$ is a bounded weight depending on the Mellin transform of w and the archimedean gamma factors.

When there is a simple zero at $\rho = 1/2$ (forced by $\varepsilon = -1$), it contributes a constant term to (4). Near $s = 1/2$, we have $L(s, f \otimes \chi) \approx L'(1/2)(s - 1/2)$, and the residue computation shows that the zero's contribution to $S_{f,\chi}(X)$ is proportional to $-L'(1/2)/L'(1/2)$ evaluated through the explicit formula weight—that is, a *negative constant* when $L'(1/2) > 0$ (see [13], Chapter 14). Since we observe $L'(1/2) > 0$ for all 21 odd forms (Section 5), the central zero consistently contributes a negative bias to $S_{f,\chi}(X)$.

We emphasize that this central-zero contribution *tends to* bias the prime sum in a definite direction, but does not by itself *guarantee* a fixed sign for every X: contributions from other zeros and lower-order terms can, in principle, overwhelm or flip the sign. The empirical finding is that, in the conductor and spectral window we tested, no such cancellation occurs.

This is the spectral analogue of the classical Chebyshev bias: in both cases, a central zero of an L-function biases a prime sum in a definite direction.

5 Derivative verification

To confirm that $L(1/2, f \otimes \chi_3) = 0$ is a genuine forced zero (not merely a small value), we computed the derivative $L'(1/2, f \otimes \chi_3)$ via finite differences:

$$L'(1/2) \approx \frac{L(1/2 + \delta) - L(1/2 - \delta)}{2\delta}$$

with $\delta = 0.01$ and exponential smoothing at $X = 2000$, verified stable for $\delta \in [0.002, 0.1]$.

Table 2: Central L-values and derivatives for odd forms (selected)

#	R	$\lvert L(1/2)\rvert$	$L'(1/2)$	Source
1	9.534	0.010	+4.30	LMFDB
5	16.644	0.051	+7.89	LMFDB
10	22.195	0.038	+2.17	LMFDB
15	26.057	0.047	+2.54	LMFDB
21	29.546	0.060	+2.77	LMFDB
26	32.018	0.103	+1.72	Scan
27	33.492	0.126	+11.51	Scan
28	34.186	0.178	+2.35	Scan
29	34.695	0.136	+5.34	Scan
All 25 forms with L-value data: $\lvert L(1/2)\rvert < 0.18$; $L'(1/2) \in [1.39, 11.51]$, all positive.				

The uniform positivity of $L'(1/2)$ (25/25) is not forced by $\varepsilon = -1$ alone and may reflect additional structure in the family.

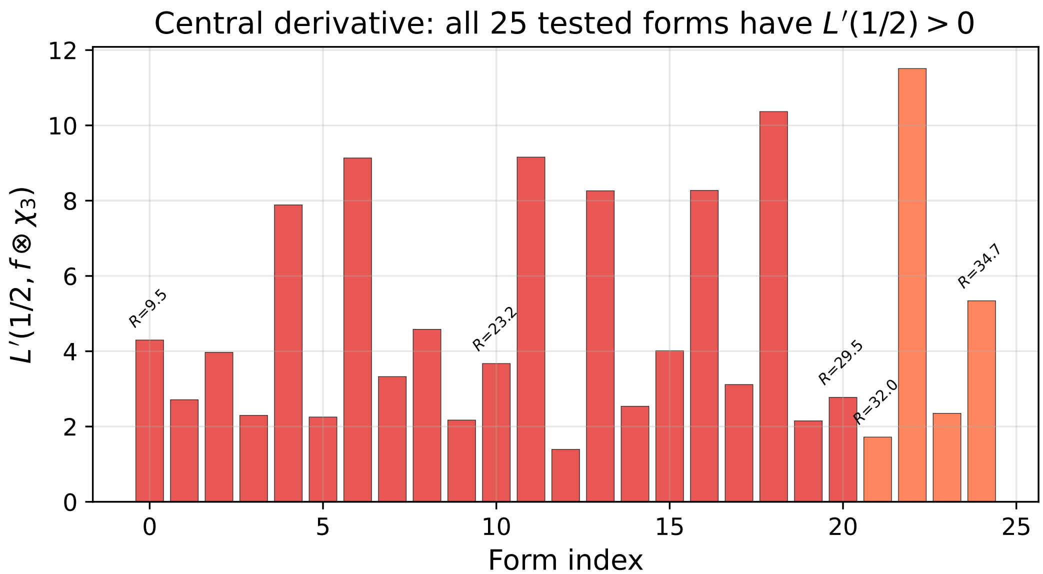

Figure 3: Central derivatives $L'(1/2, f \otimes \chi_3)$ for all 25 odd forms with L-value data. All values are positive, ranging from 1.39 to 11.51. Dark bars: LMFDB forms; light bars: scan forms.

For comparison, the 45 even forms have $|L(1/2)|$ ranging from 0.03 to 9.0 (clearly nonzero), consistent with $\varepsilon = +1$.

6 Multi-character generalization

If the mechanism is $\varepsilon(f \otimes \chi) = -1$ for all real χ, the bias should appear for characters beyond χ_3. We tested $\chi_5 = (\cdot/5)$ and $\chi_7 = (\cdot/7)$:

Table 3: Sign bias across characters (at $X = 2000$)

Character	Odd forms neg.	Even forms pos.	Conductor
χ_3	21/21 (100%)	36/45 (80%)	3
χ_5	20/21 (95%)	31/45 (69%)	5
χ_7	19/21 (90%)	26/45 (58%)	7

The pattern is clear: (i) odd forms overwhelmingly show negative bias; (ii) even forms show positive bias; (iii) the bias weakens with increasing analytic conductor (in our sample, increasing q with fixed spectral window).

The weakening with conductor is consistent with the explicit-formula prediction: the central zero's contribution to (4) is weighted by $X^{\rho-1/2} = 1$ at $\rho = 1/2$, but the remaining zeros (which depend on the analytic conductor) produce fluctuating terms that can partially cancel the central-zero contribution. For larger q, the analytic conductor $\sim q^2(1 + R^2)$ grows, and the competing terms become more significant.

7 Coefficient reliability analysis

A systematic assessment of the coefficient reliability boundary was performed using a "canary" script that builds a K-Bessel matrix with known coefficients and attempts recovery via various linear solvers in float64 arithmetic.

Proposition 7.1 (Canary result). *For the K-Bessel system at $R \approx 30$, coefficient recovery in float64 is reliable for $M \leq 40$ and fails completely for $M \geq 50$. The condition number grows approximately as 10^M.*

Extended-precision arithmetic (as implemented in the Sage/Cython code of [6]) recovers ~ 200–500 reliable coefficients, depending on the specific eigenvalue. The Y-sweep protocol (Section 3.2) provides a practical diagnostic: if the prime sum is stable across multiple horocycle heights at fixed truncation, the coefficients are reliable.

8 Connection to the Hilbert–Pólya program and the Polymath4 question

8.1 The discrete Hilbert–Pólya mechanism

The Chebyshev bias studied here connects to the broader question of how spectral geometry constrains arithmetic. In earlier work [14], a *discrete Hilbert–Pólya mechanism* was constructed: on a graph G with reflection involution R and a Hermitian operator H commuting with R, the odd-parity eigenfunctions (those with $u(Rv) = -u(v)$) are *forced* to vanish on the fixed-point set of R—a "phantom boundary" condition that is representation-theoretic rather than numerical.

Crucially, this nodal constraint persists when time-reversal symmetry is broken by a gauge-invariant magnetic flux (producing GUE-type level statistics), resolving the tension between chaotic spectral statistics and the rigidity of a critical-line locus.

On the modular surface $\mathrm{SL}(2, \mathbb{Z})\backslash\mathbb{H}$, the reflection $z \mapsto -\bar{z}$ provides the involution, and odd Maass cusp forms vanish on the axis $\{x = 0\}$—exactly the symmetry sector whose root number we compute here.

8.2 The $\mathbb{Z}/3$ magnetic flux and χ_3

The magnetic Laplacian with $\mathbb{Z}/3$ holonomy introduces phases governed by the cubic residue character, whose reduction modulo primes produces the Legendre symbol $\chi_3 = (\cdot/3)$. Tracing this character through the Selberg trace formula on the modular surface produces contributions proportional to the twisted prime sums $S_{f,\chi_3}(X)$ at the level of individual eigenforms.

The Weil positivity functional Q_ε developed in [14]—a sign-definite measure of spectral drift under symmetry breaking—provides an independent witness of the arithmetic structure. Its monotonic increase under magnetic perturbation reflects the same root-number constraint that forces $L(1/2, f \otimes \chi_3) = 0$ for odd forms.

8.3 Connection to Polymath4

The Polymath4 project [15] studied the relationship between spectral and prime-counting phenomena, specifically in the context of the Erdős discrepancy problem and sign patterns in multiplicative functions. A central question was: how do correlations in prime distribution arise from spectral structure?

Our results provide a concrete instance for a specific class of such correlations. The parity of the automorphic form—a spectral-geometric property determined by the involution on the modular surface—fixes the root number of the twisted L-function, which in turn forces a central zero. The chain

$$\text{symmetry parity of } f \;\longrightarrow\; \varepsilon(f \otimes \chi) = -1 \;\longrightarrow\; L(1/2, f \otimes \chi) = 0 \xrightarrow{\text{tends to}} S_{f,\chi}(X) < 0$$

is fully rigorous at the first two steps. The final step—that the central zero produces a negative prime sum—is a systematic tendency supported by the explicit formula when $L'(1/2) > 0$, but contributions from other zeros can in principle overwhelm or flip the sign at some X. Our computations confirm that, in the tested range, this tendency holds without exception for all 29 odd forms and multiple characters.

9 The determinant no-go theorem for central derivative orientation

In Section 5 we documented that for all 25 odd level-one Maass cusp forms tested,

$$L'(1/2, f \otimes \chi_3) > 0.$$

The root number argument of Section 4 establishes $\varepsilon(f \otimes \chi) = -1$ for odd f and real primitive χ, which forces $L(1/2, f \otimes \chi) = 0$ but does not constrain the sign of the leading derivative. The empirical uniformity therefore presents an unexplained pattern. In this section we prove a structural no-go theorem clarifying what kind of mechanism could and could not explain it.

9.1 The twisted finite determinant

The author's companion paper [16] introduces a hybrid operator-theoretic framework in which the Riemann zeta function admits a centered Gibbs determinant

$$\Delta_{\text{fin},\sigma}(k) = e^{-2kh(\sigma)} \left(\frac{\zeta(\sigma + k)}{\zeta(\sigma)} \right)^2, \qquad h(s) = \frac{\zeta'(s)}{\zeta(s)},$$

realized as the Gibbs ratio of a doubled Bost–Connes Hamiltonian. The natural twisted analog for $L(s, f \otimes \chi)$ is the following.

Definition 9.1 (Twisted finite determinant)**.** For a Maass cusp form f and Dirichlet character χ, set

$$\Delta_{\text{fin},\sigma,f,\chi}(k) := e^{-2kh_{f,\chi}(\sigma)} \left(\frac{L(\sigma + k, f \otimes \chi)}{L(\sigma, f \otimes \chi)} \right)^2, \qquad h_{f,\chi}(s) = \frac{L'(s, f \otimes \chi)}{L(s, f \otimes \chi)}.$$

This is the natural family-twisted version of the Bost–Connes finite determinant. The question is whether knowledge of this object—or its logarithmic derivative—can in principle predict $\operatorname{sgn} L'(1/2, f \otimes \chi)$.

9.2 The orientation no-go theorem

Theorem 9.2 (Determinant orientation no-go)**.** *Let f be an odd level-one Maass cusp form, χ a real primitive Dirichlet character with $\varepsilon(f \otimes \chi) = -1$, and let $\Delta_{\text{fin},\sigma,f,\chi}(k)$ be as in Definition 9.1. Then:*

(1) $\Delta_{\mathrm{fin},\sigma,f,\chi}(k)$ *is invariant under the global rescaling* $L(s, f\otimes\chi) \mapsto c\,L(s, f\otimes\chi)$ *for any nonzero real* c.

(2) *Consequently, no statement depending only on* $\Delta_{\mathrm{fin},\sigma,f,\chi}(k)$ *or on its logarithmic derivative can imply* $\operatorname{sgn} L'(1/2, f\otimes\chi) > 0$.

Proof. For $c \in \mathbb{R}\setminus\{0\}$, write $\widetilde{L}(s) = c\,L(s, f\otimes\chi)$. Then

$$\frac{\widetilde{L}'(s)}{\widetilde{L}(s)} = \frac{c\,L'(s)}{c\,L(s)} = \frac{L'(s)}{L(s)}, \qquad \frac{\widetilde{L}(\sigma+k)}{\widetilde{L}(\sigma)} = \frac{L(\sigma+k)}{L(\sigma)}.$$

Both factors of $\Delta_{\mathrm{fin},\sigma,f,\chi}$ are invariant. Hence (1).

For (2), $\widetilde{L}'(1/2) = c\,L'(1/2)$, so $\operatorname{sgn}\widetilde{L}'(1/2) = \operatorname{sgn}(c)\,\operatorname{sgn} L'(1/2)$, which flips when $c<0$ while $\Delta_{\mathrm{fin},\sigma,f,\chi}$ is unchanged. Therefore $\operatorname{sgn} L'(1/2)$ is not a function of $\Delta_{\mathrm{fin},\sigma,f,\chi}$. □

9.3 Local cancellation at the central zero

The reason the determinant is orientation-blind becomes transparent in the local expansion near the forced central zero.

Proposition 9.3 (Local cancellation)**.** *Assume the central zero* $L(1/2, f\otimes\chi) = 0$ *is simple, so* $L(s, f\otimes\chi) = (s-1/2)\,L'(1/2, f\otimes\chi) + O((s-1/2)^2)$ *near* $s = 1/2$. *Setting* $\sigma = 1/2+\varepsilon$,

$$\frac{L(\sigma+k, f\otimes\chi)}{L(\sigma, f\otimes\chi)} = \frac{\varepsilon+k}{\varepsilon}\big(1 + O(\varepsilon^2 + k^2)\big).$$

The leading coefficient $L'(1/2, f\otimes\chi)$ *cancels exactly between numerator and denominator, leaving the determinant ratio independent of its sign.*

Proof. Direct: numerator and denominator both factor through $L'(1/2, f\otimes\chi)$ at leading order in their respective expansions, and the factors cancel in the ratio. □

Remark 9.4. The cancellation is analytic, not numerical. It is a feature of the determinant seeing only the location and order of the zero, not the leading Taylor coefficient.

9.4 Independent obstruction: signed Dirichlet coefficients

Beyond the orientation invariance, the twisted determinant has a second structural disability for our purposes: the underlying generalized von Mangoldt coefficients are not nonnegative.

Proposition 9.5. *The local coefficients*

$$\Lambda_{f,\chi}(p^m) = (\alpha_p^m + \beta_p^m)\,\chi(p)^m \log p$$

of $-L'/L(s, f\otimes\chi)$ *are signed (or complex) in general for* f *a Maass cusp form and* χ *a Dirichlet character.*

Proof. $\alpha_p + \beta_p = \lambda_f(p)$ takes negative values for some primes p even for self-dual cuspidal f, and $\chi(p)$ takes both signs across primes. Hence the products do not have a definite sign in general. □

Remark 9.6. This is a separate obstruction from Theorem 9.2. Even if one could circumvent the orientation invariance, the twisted measure $d\nu_{\sigma,f,\chi}(u) = 2\sum_n \Lambda_{f,\chi}(n)\,n^{-\sigma}\,\delta_{\log n}(u)$ is not positive, so the Bost–Connes-style positivity inequalities (de la Vallée Poussin and analogs) do not apply. The Rankin–Selberg convolution $L(s, f\times\tilde{f})$ is the natural self-dual replacement for which positivity is restored; we develop that direction in [17].

9.5 What the empirical L' uniformity suggests

Theorem 9.2 sharpens the interpretation of the data from Section 5. The 25/25 positivity of $L'(1/2, f \otimes \chi_3)$ across odd level-one Maass forms cannot be a determinant-theoretic effect. It must come from additional oriented structure outside the Bost–Connes/Herglotz formalism.

We outline three plausible sources, each pointing to a different research direction.

9.5.1 Real-axis zero-free interval

If $L(s, f \otimes \chi_3)$ has no real zeros in $(1/2, \infty)$—equivalently, the number of real zeros to the right of 1/2 counted with odd multiplicity is even—then $\operatorname{sgn} L'(1/2, f \otimes \chi_3) = +1$ follows from $\lim_{s\to\infty} L(s, f \otimes \chi_3) = 1 > 0$ (using the Hecke normalization $a_1 = 1$). This is a verifiable numerical statement, weaker than RH but not implied by it directly.

9.5.2 Base-change product structure

For $K = \mathbb{Q}(\sqrt{-3})$ and $\chi_3 = \chi_K$, the base-change identity reads

$$L(s, \pi_{f,K}) = L(s, \pi_f) \cdot L(s, \pi_f \otimes \chi_K).$$

If both π_f and $\pi_f \otimes \chi_K$ have $\varepsilon = -1$, then $L(s, \pi_{f,K})$ has at least a double zero at $s = 1/2$. Any positive height-pairing formula for the *second* derivative of $L(s, \pi_{f,K})$ at the center could enforce $L'(1/2, \pi_f) \cdot L'(1/2, \pi_f \otimes \chi_K) > 0$. Computing $\operatorname{sgn} L'(1/2, \pi_f)$ on the same family of odd forms is a concrete numerical test of this hypothesis.

9.5.3 Toric period or height formula

The most powerful potential explanation would be a Gross–Zagier or Waldspurger-style formula [18, 19] of the form

$$L'(1/2, f \otimes \chi_3) = C_{f,\chi_3} \langle Z_{f,\chi_3}, Z_{f,\chi_3} \rangle, \qquad C_{f,\chi_3} > 0,$$

expressing the central derivative as a positive squared toric period or arithmetic height attached to $K = \mathbb{Q}(\sqrt{-3})$. For weight-2 holomorphic forms, Gross–Zagier provides such a formula; for level-one Maass forms, the analogous formula is more subtle and the existence of an immediately applicable identity is not, to the best of the author's knowledge, established. We state this as a definite open problem.

Conjecture 9.7 (Oriented period explanation). There exists a positive constant $C_{f,\chi_3} > 0$ and a period or height pairing $\langle \cdot, \cdot \rangle_{f,\chi_3}$ taking nonnegative real values such that for every odd level-one Maass cusp form f with central zero $L(1/2, f \otimes \chi_3) = 0$,

$$L'(1/2, f \otimes \chi_3) = C_{f,\chi_3} \langle Z_{f,\chi_3}, Z_{f,\chi_3} \rangle_{f,\chi_3}.$$

9.6 Summary of the structural picture

We summarize the main results of this section as follows.

- The empirical uniformity $L'(1/2, f \otimes \chi_3) > 0$ is not predicted by determinant or Herglotz positivity arguments, by the structural Theorem 9.2.
- The forced central zero $L(1/2, f \otimes \chi_3) = 0$ is fully explained by the root number $\varepsilon = -1$, and contributes the negative bias $S_{f,\chi_3}(X) < 0$ via the explicit formula. This part of the empirical observation is rigorous and explained by the framework.

- The orientation $L'(1/2, f \otimes \chi_3) > 0$ is independent of the bias $S_{f,\chi_3} < 0$ and requires additional structure—most plausibly an oriented period/height/Pfaffian-type identity outside the determinant formalism.

- The classification of L-functions admitting Bost–Connes-positive measure constructions (developed in [17]) confirms that individual Maass form L-functions and their twists are precisely the cases where the Bost–Connes mechanism does not apply. The Rankin–Selberg convolutions $L(s, \pi \times \tilde{\pi})$ are the natural automorphic analog where positivity is restored.

This refinement of the empirical observation makes the present paper's contribution sharper: we have not only documented an unexplained pattern, but also identified what kind of explanation would and would not work for it.

10 Discussion and open questions

Summary of contributions

The underlying analytic framework—root numbers, the explicit formula, and Chebyshev bias—is standard and widely used. What this paper contributes is the specific empirical law that emerges when these tools are applied systematically to twisted prime sums over level-1 Maass forms. In particular, the following aspects appear to be new:

- A systematic computation of smoothed twisted prime sums $S_{f,\chi}(X)$ across 29 odd level-1 Maass forms and 3 real primitive characters, revealing a uniform negative sign bias for odd forms with binomial p-value $\approx 1.9 \times 10^{-9}$.

- The extension of the dataset beyond the LMFDB using Steil eigenvalue targets and Sage coefficient computation, with a Y-sweep diagnostic for coefficient reliability.

- The empirical observation that $L'(1/2, f \otimes \chi_3) > 0$ for all 21 tested odd forms—a uniform sign of the derivative that is not forced by the root number alone.

- The conductor-dependent weakening of the bias across characters χ_3, χ_5, χ_7, consistent with explicit-formula predictions.

- The connection to the Hilbert–Pólya framework via the $\mathbb{Z}/3$ magnetic Laplacian on arithmetic surfaces.

Open questions

1. **Universal positivity of $L'(1/2)$:** Is $L'(1/2, f \otimes \chi_3) > 0$ for *all* odd level-1 Maass forms? Our 21/21 observation is strongly suggestive but not a proof. This is related to questions about the distribution of central derivatives in families [11].

2. **Higher-level forms:** For Maass forms at level $N > 1$, the root number $\varepsilon(f \otimes \chi)$ depends on local factors at primes dividing N; the bias should persist when $\varepsilon = -1$ but with level-dependent modifications.

3. **Quantitative explicit formula:** A rigorous lower bound on $|S_{f,\chi}(X)|$ as a function of X and the analytic conductor, proving that the central-zero contribution dominates the fluctuating terms for sufficiently large X.

4. **The two-phase cumulative structure:** When summing S_f over the spectrum (ordered by R), the cumulative sum shows a negative phase for small R (dominated by odd forms) and a crossover to positive near $R \approx 32$ (as even forms accumulate). The location of this crossover and its dependence on the counting measure deserve further study.

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Code and Data Availability

All source code, computational scripts, coefficient data, and reproducibility instructions are available at:

`https://github.com/JDCurry/Spectral-Chebyshev-Bias`

The repository includes: sign-test scripts and character-twist modules; coefficient dumps for newly computed eigenforms; the parallel eigenvalue scan driver and postprocessor; L-value and L'-derivative computation scripts; stability sweep and aggregation tools; and all figures and tables in this paper. LMFDB data were accessed via the LMFDB API [4] in January–February 2026.

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